# A Convergence Analysis of FedReRA

With a set of assumptions, we perform the theoretical convergence analysis of FedReRA. The proof techniques are borrowed from FedNH.

Assumption 1. *For all clients $i \in [N]$, the function $\mathcal{L}_i(\boldsymbol{\phi})$ is bounded below and L-smooth with respect to $\boldsymbol{\phi}$. That is, there exists a constant $L > 0$ such that for any $\boldsymbol{\phi}_1, \boldsymbol{\phi}_2$,*

$$\|\nabla_{\boldsymbol{\phi}}\mathcal{L}_i(\boldsymbol{\phi}_1) - \nabla_{\boldsymbol{\phi}}\mathcal{L}_i(\boldsymbol{\phi}_2)\| \le L\,\|\boldsymbol{\phi}_1 - \boldsymbol{\phi}_2\|. \tag{1}$$

Assumption 2. *Let $\boldsymbol{G}_i(\boldsymbol{\phi})$ denote the stochastic gradient of $\mathcal{L}_i$ with respect to $\boldsymbol{\phi}$, computed on a random mini-batch from the local dataset $\mathcal{D}_i$. There exists a constant $\sigma > 0$ such that, for all $i \in [N]$ and all $\boldsymbol{\phi}$,*

$$\mathbb{E}\big[\boldsymbol{G}_i(\boldsymbol{\phi})\big] = \nabla_{\boldsymbol{\phi}}\mathcal{L}_i(\boldsymbol{\phi}), \tag{2}$$

$$\mathbb{E}\big[\|\boldsymbol{G}_i(\boldsymbol{\phi}) - \nabla_{\boldsymbol{\phi}}\mathcal{L}_i(\boldsymbol{\phi})\|^2\big] \le \sigma^2. \tag{3}$$

Assumption 3. *There exist constants $M_G > 0$ and $M_f > 0$ such that $\mathbb{E}\big[\|\boldsymbol{G}_i(\boldsymbol{\phi})\|^2\big] \le M_G^2$ and $\|\nabla_{\boldsymbol{\theta}_f} f(\boldsymbol{x};\boldsymbol{\theta}_f)\| \le M_f$ for all $\boldsymbol{\phi} = (\boldsymbol{\theta}, \mathbf{W}) \in \mathcal{B}$ and $\boldsymbol{x}$, where $\mathcal{B}$ is a bounded set.*

Lemma 1. *For any client i, the partial gradients of the local objective function with respect to $\boldsymbol{\theta}$ and $\mathbf{W}$ are bounded: $\|\nabla_{\boldsymbol{\theta}}\mathcal{L}_i(\boldsymbol{\theta}, \mathbf{W})\| \le M_G$ and $\|\nabla_{\mathbf{W}}\mathcal{L}_i(\boldsymbol{\theta}, \mathbf{W})\| \le M_G$.*

Proof. By the definition of parameter $\boldsymbol{\phi} = [\boldsymbol{\theta}, \mathbf{W}]$, it holds that $\|\nabla_{\boldsymbol{\theta}}\mathcal{L}_i\| \le \|\nabla_{\boldsymbol{\phi}}\mathcal{L}_i\|$ and $\|\nabla_{\mathbf{W}}\mathcal{L}_i\| \le \|\nabla_{\boldsymbol{\phi}}\mathcal{L}_i\|$. Following Assumption 3 and using Jensen's inequality, we have:

$$\|\nabla_{\boldsymbol{\phi}}\mathcal{L}_i(\boldsymbol{\phi})\| = \|\mathbb{E}[\boldsymbol{G}_i(\boldsymbol{\phi})]\| \le \sqrt{\mathbb{E}[\|\boldsymbol{G}_i(\boldsymbol{\phi})\|^2]} \le M_G. \tag{4}$$

Substituting this into the partial gradient inequalities, we obtain $\|\nabla_{\boldsymbol{\theta}}\mathcal{L}_i\| \le M_G$ and $\|\nabla_{\mathbf{W}}\mathcal{L}_i\| \le M_G$. □

Theorem 2. *Let $\boldsymbol{\phi}^\star$ be an element sampled uniformly at random from the global iterate set $\{\boldsymbol{\phi}^t\}$ as the solution. Then for any $\epsilon > 0$, set $\rho \in (\nu_1(\epsilon), 1)$ and $\eta \in (0, \nu_2(\epsilon))$, if $R > O(\epsilon^{-1})$, one gets $\mathbb{E}\big[\|\nabla_{\boldsymbol{\phi}}\mathcal{L}(\boldsymbol{\phi}^*)\|^2\big] \le \epsilon$, where $\nu_1(\epsilon), \nu_2(\epsilon)$ are some positive constants.*

Proof. We first establish the standard sufficient decrease result, where the amount of progress can be made (in expectation) by performing $E$ steps of local stochastic gradient descent. From Assumption 1, we have:

$$\begin{aligned}\mathcal{L}_i(\boldsymbol{\phi}_i^{t,e}) &\le \mathcal{L}_i(\boldsymbol{\phi}_i^{t,e-1}) + \nabla_{\boldsymbol{\phi}}\mathcal{L}_i(\boldsymbol{\phi}_i^{t,e-1})^\top(\boldsymbol{\phi}_i^{t,e} - \boldsymbol{\phi}_i^{t,e-1}) \\ &\quad + \frac{L}{2}\|\boldsymbol{\phi}_i^{t,e} - \boldsymbol{\phi}_i^{t,e-1}\|^2 \\ &= \mathcal{L}_i(\boldsymbol{\phi}_i^{t,e-1}) - \eta\nabla_{\boldsymbol{\phi}}\mathcal{L}_i(\boldsymbol{\phi}_i^{t,e-1})^\top\boldsymbol{G}_i(\boldsymbol{\phi}_i^{t,e-1}) \\ &\quad + \frac{L\eta^2}{2}\|\boldsymbol{G}_i(\boldsymbol{\phi}_i^{t,e-1})\|^2\end{aligned} \tag{5}$$

Taking the expectation with respect to the randomness of subsampling local data $\mathcal{D}_i$ on both sides of (5) and together with Assumption 2, we obtain:

$$\begin{aligned}&\mathbb{E}[\mathcal{L}_i(\boldsymbol{\phi}_i^{t,e})] \\ &\le \mathcal{L}_i(\boldsymbol{\phi}_i^{t,e-1}) - \eta\|\nabla_{\boldsymbol{\phi}}\mathcal{L}_i(\boldsymbol{\phi}_i^{t,e-1})\|^2 \\ &\quad + \frac{L\eta^2}{2}\mathbb{E}[\|\boldsymbol{G}_i(\boldsymbol{\phi}_i^{t,e-1})\|^2] \\ &\le \mathcal{L}_i(\boldsymbol{\phi}_i^{t,e-1}) - \eta\|\nabla_{\boldsymbol{\phi}}\mathcal{L}_i(\boldsymbol{\phi}_i^{t,e-1})\|^2 \\ &\quad + \frac{L\eta^2}{2}(\|\nabla_{\boldsymbol{\phi}}\mathcal{L}_i(\boldsymbol{\phi}_i^{t,e-1})\|^2 + \sigma^2) \\ &= \mathcal{L}_i(\boldsymbol{\phi}_i^{t,e-1}) - \left(\eta - \frac{L\eta^2}{2}\right)\|\nabla_{\boldsymbol{\phi}}\mathcal{L}_i(\boldsymbol{\phi}_i^{t,e-1})\|^2 + \frac{L\eta^2\sigma^2}{2},\end{aligned} \tag{6}$$

where the second inequality utilizes the fact that $\mathbb{E}[\|X\|^2] = \mathbb{E}[\|X - \mathbb{E}[X]\|^2] + \|\mathbb{E}[X]\|^2$. By summing the above result over the local update index $e = 1, \ldots, E$, we reach the cumulative descent after one round of local training:

$$\begin{aligned}\mathbb{E}[\mathcal{L}_i(\boldsymbol{\phi}_i^{t,E})] &\le \mathcal{L}_i(\boldsymbol{\phi}_i^{t,0}) + \frac{LE\eta^2\sigma^2}{2} \\ &\quad - \left(\eta - \frac{L\eta^2}{2}\right)\sum_{e=1}^{E}\|\nabla_{\boldsymbol{\phi}}\mathcal{L}_i(\boldsymbol{\phi}_i^{t,e-1})\|^2.\end{aligned} \tag{7}$$

Next, we bound $\mathcal{L}_i(\boldsymbol{\phi}_i^{t,0}) - \mathcal{L}_i(\boldsymbol{\phi}_i^{t-1,E})$, which characterize the difference the initial state of round $t$ and the final state of round $t-1$ for client $i$.

$$\begin{aligned}&\mathcal{L}_i(\boldsymbol{\phi}_i^{t,0}) - \mathcal{L}_i(\boldsymbol{\phi}_i^{t-1,E}) \\ &= \mathcal{L}_i(\boldsymbol{\theta}_i^{t,0}, \mathbf{W}_i^{t,0}) - \mathcal{L}_i(\boldsymbol{\theta}_i^{t-1,E}, \mathbf{W}_i^{t-1,E}) \\ &= \mathcal{L}_i(\boldsymbol{\theta}_i^{t,0}, \mathbf{W}_i^{t,0}) - \mathcal{L}_i(\boldsymbol{\theta}_i^{t-1,E}, \mathbf{W}_i^{t,0}) \\ &\quad + \mathcal{L}_i(\boldsymbol{\theta}_i^{t-1,E}, \mathbf{W}_i^{t,0}) - \mathcal{L}_i(\boldsymbol{\theta}_i^{t-1,E}, \mathbf{W}_i^{t-1,E}) \\ &\le |\mathcal{L}_i(\boldsymbol{\theta}_i^{t,0}, \mathbf{W}_i^{t,0}) - \mathcal{L}_i(\boldsymbol{\theta}_i^{t-1,E}, \mathbf{W}_i^{t,0})| \\ &\quad + |\mathcal{L}_i(\boldsymbol{\theta}_i^{t-1,E}, \mathbf{W}_i^{t,0}) - \mathcal{L}_i(\boldsymbol{\theta}_i^{t-1,E}, \mathbf{W}_i^{t-1,E})|.\end{aligned} \tag{8}$$

**We first bound** $|\mathcal{L}_i(\boldsymbol{\theta}_i^{t,0}, \mathbf{W}_i^{t,0}) - \mathcal{L}_i(\boldsymbol{\theta}_i^{t-1,E}, \mathbf{W}_i^{t,0})|$ in Eq. (8). It follows from Lemma 1 that the local objective function $\mathcal{L}_i(\boldsymbol{\theta}, \mathbf{W})$ is $M_G$-Lipschitz continuous with respect to its first argument $\boldsymbol{\theta}$. Therefore, we have

$$|\mathcal{L}_i(\boldsymbol{\theta}_i^{t,0}, \mathbf{W}_i^{t,0}) - \mathcal{L}_i(\boldsymbol{\theta}_i^{t-1,E}, \mathbf{W}_i^{t,0})| \le M_G\|\boldsymbol{\theta}_i^{t,0} - \boldsymbol{\theta}_i^{t-1,E}\|. \tag{9}$$

Since $\boldsymbol{\theta}_i^{t,0} = \frac{1}{N}\sum_{k=1}^{N}\boldsymbol{\theta}_k^{t-1,E}$, using the same technique from the Lemma 1 in [33], one has

$$\mathbb{E}[\|\boldsymbol{\theta}_i^{t,0} - \boldsymbol{\theta}_i^{t-1,E}\|] \le 2\eta E M_G. \tag{10}$$

Combining (9) and (10) and taking expectation, we have

$$\mathbb{E}[|\mathcal{L}_i(\boldsymbol{\theta}_i^{t,0}, \mathbf{W}_i^{t,0}) - \mathcal{L}_i(\boldsymbol{\theta}_i^{t-1,E}, \mathbf{W}_i^{t,0})|] \le 2\eta E M_G^2. \tag{11}$$

**We then bound** $|\mathcal{L}_i(\boldsymbol{\theta}_i^{t-1,E}, \mathbf{W}_i^{t,0}) - \mathcal{L}_i(\boldsymbol{\theta}_i^{t-1,E}, \mathbf{W}_i^{t-1,E})|$ in Eq. (8). Similar to $\boldsymbol{\theta}$, $\mathcal{L}_i(\boldsymbol{\theta}, \mathbf{W})$ is also $M_G$-Lipschitz continuous with respect to the auxiliary classifier $\mathbf{W}$. Therefore, we have

$$\begin{aligned}&|\mathcal{L}_i(\boldsymbol{\theta}_i^{t-1,E}, \mathbf{W}_i^{t,0}) - \mathcal{L}_i(\boldsymbol{\theta}_i^{t-1,E}, \mathbf{W}_i^{t-1,E})| \\ &\le M_G\|\mathbf{W}_i^{t,0} - \mathbf{W}_i^{t-1,E}\| \\ &\le M_G(\|\mathbf{W}_i^{t,0} - \mathbf{W}_i^{t-1,0}\| + \|\mathbf{W}_i^{t-1,0} - \mathbf{W}_i^{t-1,E}\|),\end{aligned} \tag{12}$$

We first bound the second term in Eq. (12). By the definition of the local update rule for $\mathbf{W}$, we have

$$\mathbf{W}_i^{t-1,E} - \mathbf{W}_i^{t-1,0} = -\eta \sum_{e=1}^{E} G_i{}^{(\mathbf{W})}(\boldsymbol{\theta}_i^{t-1,e-1}, \mathbf{W}_i^{t-1,e-1}), \tag{13}$$

where $\boldsymbol{G}_i^{\mathbf{W}}$ denotes the stochastic gradient with respect to $\mathbf{W}$. Taking the norm and expectation on both sides, we have

$$\begin{aligned}
&\mathbb{E}[\|\mathbf{W}_i^{t-1,E} - \mathbf{W}i^{t-1,0}\|] \\
&= \eta \mathbb{E}[\| \sum_{e=1}^{E} G_i^{(\mathbf{W})}(\boldsymbol{\theta}_i^{t-1,e-1}, \mathbf{W}_i^{t-1,e-1})\|] \\
&\le \eta \sum_{e=1}^{E} \mathbb{E}[\|G_i^{(\mathbf{W})}(\boldsymbol{\theta}_i^{t-1,e-1}, \mathbf{W}_i^{t-1,e-1})\|] \\
&\le \eta E M_G,
\end{aligned} \tag{14}$$

We then bound the first term in Eq. (12). From the update rule in FedReRA, the global auxiliary classifier is updated by aggregating relative prototypes from clients. We have

$$\begin{aligned}
\|\mathbf{W}_i^{t,0} - \mathbf{W}_i^{t-1,0}\| &\le \frac{1}{N} \sum_{k=1}^{N} \sum_{c=1}^{|C|} ((1-\rho)\|\hat{\boldsymbol{t}}_{k,c}^{t} - \hat{\boldsymbol{t}}_{k,c}^{t-1}\| \\
&\quad + \rho \|\mathbf{W}_{i,c}^{t-1,0} - \mathbf{W}_{i,c}^{t-2,0}\|) \\
&\le |C| \cdot \frac{1}{N} \sum_{k=1}^{N} (1-\rho) \|\hat{\boldsymbol{t}}_{k,c}^{t} - \hat{\boldsymbol{t}}_{k,c}^{t-1}\|.
\end{aligned} \tag{15}$$

By the definition of relative prototype and Assumption 3, we have

$$\begin{aligned}
&\|\hat{\boldsymbol{t}}_{k,c}^{t} - \hat{\boldsymbol{t}}_{k,c}^{t-1}\| \\
&\le \|\boldsymbol{t}_{k,c}^{t} - \boldsymbol{t}_{k,c}^{t-1}\| + \|\boldsymbol{a}_k^{t} - \boldsymbol{a}_k^{t-1}\| \\
&= \|\frac{1}{|\mathcal{D}_{k,c}|} \sum_{x \in \mathcal{D}_{k,c}} (f(x; \boldsymbol{\theta}^{f^{t,E}}_{k}) - f(x; \boldsymbol{\theta}^{f^{t-1,E}}_{k}))\| \\
&\quad + \|\frac{1}{|\mathcal{D}_p|} \sum_{x \in \mathcal{D}_p} (f(x; \boldsymbol{\theta}^{f^{t,E}}_{k}) - f(x; \boldsymbol{\theta}^{f^{t-1,E}}_{k}))\| \\
&\le \frac{1}{|\mathcal{D}_{k,c}|} \sum_{x \in \mathcal{D}_{k,c}} M_f \|\boldsymbol{\theta}^{f^{t,E}}_{k} - \boldsymbol{\theta}^{f^{t-1,E}}_{k}\| \\
&\quad + \frac{1}{|\mathcal{D}_p|} \sum_{x \in \mathcal{D}_p} M_f \|\boldsymbol{\theta}^{f^{t,E}}_{k} - \boldsymbol{\theta}^{f^{t-1,E}}_{k}\| \\
&= 2M_f \|\boldsymbol{\theta}^{f^{t,E}}_{k} - \boldsymbol{\theta}^{f^{t-1,E}}_{k}\|.
\end{aligned} \tag{16}$$

By splitting the parameter difference into local update and global aggregation, we have

$$\begin{aligned}
&\|\boldsymbol{\theta}^{f^{t,E}}_{k} - \boldsymbol{\theta}^{f^{t-1,E}}_{k}\| \\
&\le \sum_{e=1}^{E} \|\boldsymbol{\theta}^{f^{t,e}}_{k} - \boldsymbol{\theta}^{f^{t,e-1}}_{k}\| + \|\boldsymbol{\theta}^{f^{t,0}}_{k} - \boldsymbol{\theta}^{f^{t-1,E}}_{k}\| \\
&= \eta \sum_{e=1}^{E} \|G_i^{\theta^f}(\boldsymbol{\phi}_k^{t,e-1})\| + \|\boldsymbol{\theta}^{f^{t,0}}_{k} - \boldsymbol{\theta}^{f^{t-1,E}}_{k}\|.
\end{aligned} \tag{17}$$

Taking the expectation on both sides and together with (10), we have

$$\mathbb{E}[\|\boldsymbol{\theta}_k^{t,E} - \boldsymbol{\theta}_k^{t-1,E}\|] \le \eta E M_G + 2\eta E M_G = 3\eta E M_G. \tag{18}$$

From (12) to (18), we have

$$\begin{aligned}
&\mathbb{E}[|\mathcal{L}_i(\boldsymbol{\theta}_i^{t-1,E}, \mathbf{W}_i^{t,0}) - \mathcal{L}_i(\boldsymbol{\theta}_i^{t-1,E}, \mathbf{W}_i^{t-1,E})|] \\
&\le M_G(\mathbb{E}[\|\mathbf{W}_i^{t,0} - \mathbf{W}_i^{t-1,0}\|] + \mathbb{E}[\|\mathbf{W}_i^{t-1,0} - \mathbf{W}_i^{t-1,E}\|]) \\
&\le M_G(6|C|(1-\rho)M_f \eta E M_G + \eta E M_G) \\
&= \eta E M_G^2 + 6|C|(1-\rho)M_f \eta E M_G^2.
\end{aligned} \tag{19}$$

Combining (8), (11) and (19) we have

$$\begin{aligned}
&\mathbb{E}[\mathcal{L}_i(\boldsymbol{\phi}_i^{t,0}) - \mathcal{L}_i(\boldsymbol{\phi}_i^{t-1,E})] \\
&\le 2\eta E M_G^2 + \eta E M_G^2 + 6(1-\rho) M_G^2 M_f \eta E |C| \\
&= 3\eta E M_G^2 + 6(1-\rho) M_G^2 M_f \eta E |C| \\
&= (1-\rho)\gamma\eta,
\end{aligned} \tag{20}$$

where $\gamma = 3EM_G M_\iota + 6M_G^2 M_f E|C|$ and $M_\iota = \frac{M_G}{1-\rho}$. By shifting the round index in (20) from $t$ to $t+1$, and combining with (20), we have

$$\begin{aligned}
&\mathbb{E}[\mathcal{L}_i(\boldsymbol{\phi}^{t}) - \mathcal{L}_i(\boldsymbol{\phi}^{t-1})] \\
&= \mathbb{E}[\mathcal{L}_i(\boldsymbol{\phi}_i^{t+1,0}) - \mathcal{L}_i(\boldsymbol{\phi}_i^{t,0})] \\
&\le -\left(\eta - \frac{L\eta^2}{2}\right) \sum_{e=1}^{E} \mathbb{E}[\|\nabla_{\boldsymbol{\phi}} \mathcal{L}_i(\boldsymbol{\phi}_i^{t,e-1})\|^2] \\
&\quad + \frac{LE\eta^2\sigma^2}{2} + (1-\rho)\gamma\eta,
\end{aligned} \tag{21}$$

which establishes a bound for the cross-round variation of the local loss evaluated at the global model states. By taking the weighted average of (21) over all clients and according to the definition of global objective function, we have

$$\begin{aligned}
&\mathbb{E}[\mathcal{L}(\boldsymbol{\phi}^{t}) - \mathcal{L}(\boldsymbol{\phi}^{t-1})] \\
&= \sum_{i=1}^{N} p_i \mathbb{E}[\mathcal{L}_i(\boldsymbol{\phi}^{t}) - \mathcal{L}_i(\boldsymbol{\phi}^{t-1})] \\
&\le -\left(\eta - \frac{L\eta^2}{2}\right) \sum_{i=1}^{N} p_i \sum_{e=1}^{E} \mathbb{E}[\|\nabla_{\boldsymbol{\phi}} \mathcal{L}_i(\boldsymbol{\phi}_i^{t,e-1})\|^2] \\
&\quad + \frac{LE\eta^2\sigma^2}{2} + (1-\rho)\gamma\eta.
\end{aligned} \tag{22}$$

Taking the initial model as a reference point, we have

$$\begin{aligned}
&\mathbb{E}[\|\nabla_{\boldsymbol{\phi}} \mathcal{L}_i(\boldsymbol{\phi}_i^{t,e-1})\|^2] \\
&\ge \frac{1}{2}\|\nabla_{\boldsymbol{\phi}} \mathcal{L}_i(\boldsymbol{\phi}^{t,0})\|^2 - \mathbb{E}[\|\nabla_{\boldsymbol{\phi}} \mathcal{L}_i(\boldsymbol{\phi}_i^{t,e-1}) - \nabla_{\boldsymbol{\phi}} \mathcal{L}_i(\boldsymbol{\phi}_i^{t,0})\|^2] \\
&\ge \frac{1}{2}\|\nabla_{\boldsymbol{\phi}} \mathcal{L}_i(\boldsymbol{\phi}^{t,0})\|^2 - L^2 \mathbb{E}[\|\boldsymbol{\phi}_i^{t,e-1} - \boldsymbol{\phi}^{t,0}\|^2], \\
&\ge \frac{1}{2}\|\nabla_{\boldsymbol{\phi}} \mathcal{L}_i(\boldsymbol{\phi}^{t,0})\|^2 - L^2\eta^2(e-1)^2 M_g^2.
\end{aligned} \tag{23}$$

where the first inequality uses the vector norm inequality $\|\boldsymbol{a}\|^2 \ge \frac{1}{2}\|\boldsymbol{b}\|^2 - \|\boldsymbol{a}-\boldsymbol{b}\|^2$, the second inequality utilizes Assumption 1, and the last inequality utilizes Assumption 3. Summing over $e = 1, \ldots, E$, we have

$$\sum_{e=1}^{E} \mathbb{E}[\|\nabla_{\boldsymbol{\phi}} \mathcal{L}_i(\boldsymbol{\phi}_i^{t,e-1})\|^2] \ge \frac{E}{2}\|\nabla_{\boldsymbol{\phi}} \mathcal{L}_i(\boldsymbol{\phi}^{t})\|^2 - \frac{L^2\eta^2E^3M_G^2}{3}. \tag{24}$$

Taking weighted average over all clients, we have

$$\begin{aligned}&\sum_{i=1}^{N} p_i \sum_{e=1}^{E} \mathbb{E}[\|\nabla_{\boldsymbol{\phi}} \mathcal{L}_i(\boldsymbol{\phi}_i^{t,e-1})\|^2] \\ &\geq \frac{E}{2} \sum_{i=1}^{N} p_i \|\nabla_{\boldsymbol{\phi}} \mathcal{L}_i(\boldsymbol{\phi}^t)\|^2 - \frac{L^2\eta^2 E^3 M_G^2}{3} \\ &\geq \frac{E}{2} \mathbb{E}[\|\nabla_{\boldsymbol{\phi}} \mathcal{L}(\boldsymbol{\phi}^t)\|^2] - \frac{L^2\eta^2 E^3 M_G^2}{3},\end{aligned} \tag{25}$$

where the second inequality applies Jensen's Inequality. Combining (25) and (22), and let $0 < \eta < \frac{2}{L}$, we have

$$\begin{aligned}&\mathbb{E}[\mathcal{L}(\boldsymbol{\phi}^t) - \mathcal{L}(\boldsymbol{\phi}^{t-1})] \\ &\leq -\frac{E}{2}\left(\eta - \frac{L\eta^2}{2}\right) \mathbb{E}[\|\nabla_{\boldsymbol{\phi}} \mathcal{L}(\boldsymbol{\phi}^t)\|^2] \\ &\quad + \frac{L^2\eta^2 E^3 M_G^2}{3}\left(\eta - \frac{L\eta^2}{2}\right) + \frac{LE\eta^2\sigma^2}{2} + (1-\rho)\gamma\eta,\end{aligned} \tag{26}$$

which can be arranged as

$$\begin{aligned}&\mathbb{E}[\|\nabla \mathcal{L}(\boldsymbol{\phi}^t)\|^2] \\ &\leq \frac{\mathbb{E}[\mathcal{L}(\boldsymbol{\phi}^{t-1}) - \mathcal{L}(\boldsymbol{\phi}^t)]}{\frac{E}{2}(\eta - \frac{L\eta^2}{2})} \\ &\quad + \frac{\frac{L^2\eta^2E^3M_G^2}{3}\left(\eta - \frac{L\eta^2}{2}\right) + \frac{LE\eta^2\sigma^2}{2} + (1-\rho)\gamma\eta}{\frac{E}{2}\left(\eta - \frac{L\eta^2}{2}\right)}.\end{aligned} \tag{27}$$

Taking the average expectation across all rounds, we have

$$\begin{aligned}&\frac{1}{R}\sum_{t=0}^{R-1} \mathbb{E}[\|\nabla \mathcal{L}(\boldsymbol{\phi}^t)\|^2] \\ &\leq \frac{\mathcal{L}(\boldsymbol{\phi}^0) - \mathcal{L}(\boldsymbol{\phi}^*)}{\frac{RE}{2}(\eta - \frac{L\eta^2}{2})} \\ &\quad + \frac{\frac{L^2\eta^2E^3M_G^2}{3}\left(\eta - \frac{L\eta^2}{2}\right) + \frac{LE\eta^2\sigma^2}{2} + (1-\rho)\gamma\eta}{\frac{E}{2}\left(\eta - \frac{L\eta^2}{2}\right)}.\end{aligned} \tag{28}$$

Then taking expectation with respect to the $R$, we have

$$\begin{aligned}&\mathbb{E}[\|\nabla \mathcal{L}(\boldsymbol{\phi}^*)\|^2] \\ &\leq \frac{2(\mathcal{L}(\boldsymbol{\phi}^0) - \mathcal{L}(\boldsymbol{\phi}^*))}{RE(\eta - \frac{L\eta^2}{2})} + \frac{2L^2\eta^2E^2M_G^2}{3} + \frac{L\eta\sigma^2 + \frac{2(1-\rho)\gamma}{E}}{(1 - \frac{L\eta}{2})}.\end{aligned} \tag{29}$$

To ensure that the global model converges to an $\epsilon$-stationary point such that $\mathbb{E}[\|\nabla \mathcal{L}(\boldsymbol{\phi}^*)\|^2] \leq \epsilon$ as $R$ increases, we require

$$\frac{2L^2\eta^2E^2M_G^2}{3} + \frac{L\eta\sigma^2 + \frac{2(1-\rho)\gamma}{E}}{(1 - \frac{L\eta}{2})} \leq \epsilon, \tag{30}$$

which implies the cubic function $g(\eta) = A\eta^3 + B\eta^2 + C\eta + D \leq 0$, where $A = -\frac{L^3E^2M_G^2}{3}$, $B = \frac{2L^2E^2M_G^2}{3}$, $C = L\sigma^2 + \frac{\epsilon L}{2}$, and $D = \frac{2(1-\rho)\gamma}{E} - \epsilon$ are coefficients. When $\rho \in (1 - \frac{\epsilon E}{2\gamma}, 1)$, it holds that $g(0) < 0$. Since $g(\eta)$ is a continuous polynomial with respect to $\eta$, it follows that there must exist a positive constant $\nu(\eta, L, E, M_G, \sigma^2, \rho, \gamma) > 0$ such that for any learning rate that satisfies $\eta \in (0, \min(\nu, \frac{2}{L}))$, the condition $g(\eta) < 0$ remains satisfied. Therefore, if

$$R \geq \frac{4(F(\boldsymbol{\phi}^0) - F(\boldsymbol{\phi}^*))}{\eta[\epsilon E(2 - L\eta) - \frac{2L^2\eta^2E^3M_G^2}{3}(2 - L\eta) - 2LE\eta\sigma^2 - 4(1-\rho)\gamma]}, \tag{31}$$

we reach $\mathbb{E}[\|\nabla \mathcal{L}(\boldsymbol{\phi}^*)\|^2] \leq \epsilon$. □

## B Detailed Procedure

More details about process of RKE and REI are described in Algorithm 1 and Algorithm 2, respectively.

**Algorithm 1** Training process of RKE in FedReRA

**Input:** $N$ clients with data $\{\mathcal{D}_1, \cdots, \mathcal{D}_N\}$, proxy dataset $\mathcal{D}_p$, initial model $\boldsymbol{\theta}^0$, initial auxiliary classifier $\mathbf{W}^0$, communication round $R$, learning rate $\eta$, local epochs $E$, sharing threshold $\kappa$, communication threshold $\sigma$, smooth parameter $\rho$.
**Output:** global model $\boldsymbol{\theta}$, global auxiliary classifier $\mathbf{W}$ and translation memory $\boldsymbol{a}$ of global model.

Server initializes $M^0 \leftarrow \{m^0_{ij}\}_{i,j\in[|D_p|]}$ by Equation 20.
Server initializes $\Delta \leftarrow Infinity$.
**for** each round $r = 1, \cdots, R$ **do**
  Server sends $\{\boldsymbol{\theta}^{r-1}, \mathbf{W}^{r-1}\}$ to all clients.
  *// Local Training for Clients*
  **for** each client $i$ **in parallel do**
    Initialize $\{\boldsymbol{\theta}^{r-1}_i, \mathbf{W}^{r-1}_i\} \leftarrow \{\boldsymbol{\theta}^{r-1}, \mathbf{W}^{r-1}\}$.
    Obtain $\{\boldsymbol{\theta}^r_i, \mathbf{W}^r_i\}$ by minimizing $\mathcal{L}_i$ in Equation 8 with SGD for $E$ epochs.
    Upload $\{\boldsymbol{\theta}^r_i\}$ to server.
    **if** $\Delta > \sigma$ **then**
      Initialize prototype set $\mathcal{T}^r_i \leftarrow \emptyset$
      **for** each class $c \in C_i$ with $|\mathcal{D}_{i,c}| \geq \kappa$ **do**
        Calculate absolute prototype $\boldsymbol{t}^r_{i,c}$.
        $\mathcal{T}^r_i \leftarrow \mathcal{T}^r_i \cup \{\boldsymbol{t}^r_{i,c}\}$
      **end for**
      Upload $\{\mathcal{T}^r_i\}$ to server.
    **end if**
  **end for**
  *// Global Aggregation for Server*
  Aggregate model parameters $\boldsymbol{\theta}^r$.
  Compute $\boldsymbol{a}^r \leftarrow \frac{1}{|D_p|}\sum_{\boldsymbol{x}\in D_p} f(\boldsymbol{x}; \boldsymbol{\theta}^{f,r})$.
  Compute $M^r \leftarrow \{m^r_{ij}\}_{i,j\in[|D_p|]}$ by Equation 20.
  **if** $\Delta > \sigma$ **then**
    **for** each client $i$ **do**
      Obtain $\boldsymbol{a}^r_i \leftarrow \frac{1}{|D_p|}\sum_{\boldsymbol{x}\in D_p} f(\boldsymbol{x}; \boldsymbol{\theta}^{f,r}_i)$
    **end for**
    **for** each class $c$ **do**
      Get auxiliary classifier weight $\mathbf{W}^r_c$ by Equation 9.
    **end for**
    $\Delta \leftarrow MAD(M^r, M^{r-1})$
  **else**
    $\mathbf{W}^r \leftarrow \mathbf{W}^{r-1}$
    $\Delta \leftarrow \Delta + MAD(M^r, M^{r-1})$
  **end if**
**end for**
**return** $\boldsymbol{\theta}^R, \mathbf{W}^R, \boldsymbol{a}^R$

**Algorithm 2** Inference process of REI in FedReRA

**Input:** model $\boldsymbol{\theta} = [\boldsymbol{\theta}^f, \boldsymbol{\theta}^h]$, auxiliary classifier $\mathbf{W}$, model's debiasing memory $\boldsymbol{a}$, query data $\boldsymbol{x}$.
**Output:** probability distribution $\tilde{y}$.

Branched prediction $\tilde{y}_b \leftarrow \mathrm{softmax}(\mathbf{W}(f(\boldsymbol{x}; \boldsymbol{\theta}^f) - \boldsymbol{a}))$
Original prediction $\tilde{y}_o \leftarrow \mathrm{softmax}(h(f(\boldsymbol{x}; \boldsymbol{\theta}^f); \boldsymbol{\theta}^h))$
Prediction result $\tilde{y} \leftarrow \arg\max_c \frac{1}{2}(\tilde{y}_b + \tilde{y}_o)$
**return** $\tilde{y}$